

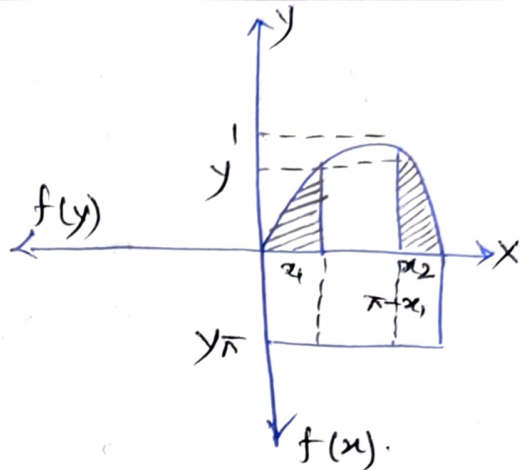
1

$$y = \sin x$$

$$x = \sin^{-1} y$$

$$x_1 = \sin^{-1} y$$

$$x_2 = \pi - \sin^{-1} y$$



$$F_y(y) = P(y \leq y) = P(\sin x \leq y)$$

$$= P(x \leq \sin^{-1} y)$$

$$\underbrace{\hspace{10em}}_{x_1 \quad x_2}$$

$$= P(x \leq x_1) + P(\pi - x_1 \leq x \leq \pi)$$

$$= P(x \leq \sin^{-1} y) + P(\pi - \sin^{-1} y < x < \pi)$$

$$= \int_0^{\sin^{-1} y} f_x(x) dx + \int_0^{\pi} f_x(x) dx - \int_0^{\pi - \sin^{-1} y} f_x(x) dx$$

$$= \frac{x}{\pi} \Big|_0^{\sin^{-1} y} + \frac{x}{\pi} \Big|_0^{\pi} - \frac{x}{\pi} \Big|_0^{\pi - \sin^{-1} y}$$

$$f_y(y) = \frac{1}{\pi} (\sin^{-1} y - 0) + \frac{1}{\pi} (\pi - 0) - \frac{1}{\pi} (\pi - \sin^{-1} y - 0)$$

$$= \frac{1}{\pi} \sin^{-1} y * 2$$

$$= \frac{2}{\pi} \sin^{-1} y = \frac{2}{\pi} x$$

$$f_y(y) = \frac{d}{dy} F_y(y) = \frac{d}{dy} \left(\frac{2}{\pi} \sin^{-1} y \right).$$

$$= \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

$$= \frac{2}{\pi} \cdot \frac{1}{\cos x}$$

$$\left| \begin{array}{l} y = \sin x \\ 1 - y^2 = \cos^2 x \end{array} \right.$$

[2]

$x, y \rightarrow$ Statistically ind.

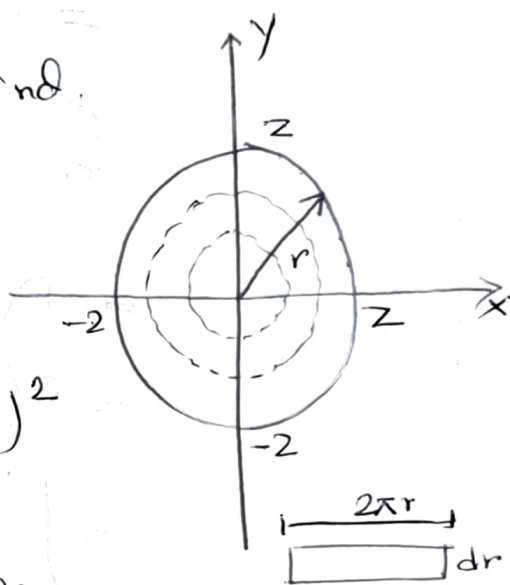
$$f(x) \rightarrow N(0, \sigma^2)$$

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2} \left(\frac{x-0}{\sigma} \right)^2}$$

$$= \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2} \cdot \frac{x^2}{\sigma^2}}$$

$$f(y) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2} \cdot \frac{y^2}{\sigma^2}}$$

$$\begin{aligned} f(x, y) \cdot dx \cdot dy &= f(x) \cdot f(y) = \frac{1}{2\pi\sigma^2} \cdot e^{-\frac{1}{2} \left(\frac{x^2}{\sigma^2} + \frac{y^2}{\sigma^2} \right)} \\ &= \frac{1}{2\pi\sigma^2} \cdot e^{-\frac{1}{2} \left(\frac{x^2 + y^2}{\sigma^2} \right)}. \end{aligned}$$



$$f(x,y) = \iint f(x,y) \cdot dx \cdot dy \rightarrow dx \cdot dy = 2\pi r dr$$

$$r^2 = x^2 + y^2$$

$$= \int \frac{1}{2\pi a^2} \cdot e^{-\frac{r^2}{2a^2}} \cdot 2\pi r dr$$

$$f(x,y) = \int_0^2 \frac{1}{a^2} \cdot e^{-\frac{r^2}{2a^2}} \cdot r dr$$

$$f(x,y) = -\frac{1}{2} \cdot e^{-\frac{r^2}{2a^2}} \Big|_0^2$$

$$= -\left(e^{-\frac{2^2}{2a^2}} - 1\right) = 1 - e^{-\frac{2^2}{2a^2}}$$

$$f(z) \cdot dz = f(x,y) \cdot dx \cdot dy$$

$$= \frac{1}{2\pi a^2} \cdot e^{-\frac{1}{2a^2}(x^2+y^2)}$$

$$f(z) = 1 - e^{-\frac{z^2}{2a^2}}$$

$$= 1 - e^{-\frac{1}{2a^2}(x^2+y^2)}$$

$$\begin{aligned} \underline{\underline{3}}] a. E[(x-c)^2] &= E[x^2 - 2xc + c^2] \\ &= E(x^2) - 2cE(x) + c^2. \end{aligned}$$

To find c that minimize $E[(x-c)^2]$.

$$\frac{d}{dc} E[(x-c)^2] = 0.$$

$$0 - 2E(x) + 2c = 0. \quad c = E(x) = \mu_x.$$

b. find c that minimizes $E[|x-c|]$.

$$E[|x-c|] = E(x-c) + E(-x+c)$$

for $x > c$ for $x < c$

$$= \int_c^\infty (x-c) f(x) dx + \int_{-\infty}^c (c-x) f(x) dx.$$

$$\begin{aligned} &= \int_c^\infty x f(x) dx + \int_{-\infty}^c x f(x) dx - \int_c^\infty c f(x) dx \\ &\quad + \int_{-\infty}^c c f(x) dx \end{aligned}$$

$$\frac{d}{dc} [|x-c|] = \int_{-\infty}^c f(x) dx - \int_c^{\infty} f(x) \cdot dx = 0.$$

$$@ c \rightarrow F(x) = 0.5 \rightarrow c \text{ is median}$$

$$\underline{\underline{c}} \quad E \left[\left\{ (y - \mu_y) - c (x - \mu_x) \right\}^n \right].$$

$$= E \left[(y - \mu_y)^2 + c^n (x - \mu_x)^n - 2c (y - \mu_y) (x - \mu_x) \right].$$

$$= E \left[(y - \mu_y)^n \right] + c^n \cdot E \left[(x - \mu_x)^n \right] - 2c \cdot E \left[(y - \mu_y) (x - \mu_x) \right].$$

$$\text{to find } c \rightarrow \frac{d}{dc} \cdot E = 0.$$

$$0 * 2c * E \left[(x - \mu_x)^n \right] - 2E \left[(y - \mu_y) (x - \mu_x) \right] = 0.$$

$$c = \frac{E \left[(y - \mu_y) (x - \mu_x) \right]}{E \left[(x - \mu_x)^2 \right]} = \frac{\text{Covariance}(x, y)}{\sigma_x^2}$$

$$4] \quad \mu = 110 \text{ ft/sec}$$

$$\sigma = 25 \text{ ft/sec}$$

$$\lambda x = \ln \frac{\mu}{\sqrt{1 + (\sigma/\mu)^2}} = \ln \left[\frac{110}{\sqrt{1 + (25/110)^2}} \right] = 4.675$$

$$\xi^2 = \ln \left(1 + \left(\frac{\sigma}{\mu} \right)^2 \right) = \ln \left(1 + \left(\frac{25}{110} \right)^2 \right) \\ = 0.0504$$

$$\xi = 0.224$$

$$P(x > 140 \text{ ft/sec}) = P(\ln x > \ln 140) = 1 - P(\ln x \leq \ln 140) \\ = 1 - P\left(2 \leq \frac{\ln 140 - 4.675}{0.224}\right) \\ = 1 - P(2 \leq 1.19)$$

$$P(x > 140 \text{ ft/sec}) = 1 - 0.883 \\ = 0.117$$

5]

Test A $\rightarrow \mu = 21, \sigma = 6.$

$$A = N(21, 6)$$

Test S $\rightarrow \mu = 1050, \sigma = 210$

$$S = N(1050, 210)$$

$$a) P(S < 1200) = ?$$

$$Z = \frac{x - \mu}{\sigma} = \frac{1200 - 1050}{210} = 0.714.$$

$$P(Z < 0.714) = 0.7611.$$

$$b) Z = 0.714 \rightarrow 0.714 = \frac{x - 21}{6} \rightarrow x = 25.284.$$

$$c) P(A > 36) = 1 - P(A \leq 36) = 1 - P\left(Z \leq \frac{36 - 21}{6}\right).$$

$$= 1 - P(Z < 2.5)$$

$$= 1 - 0.9938$$

$$= 0.0062.$$

$$P(A > 36) = 0.0062$$